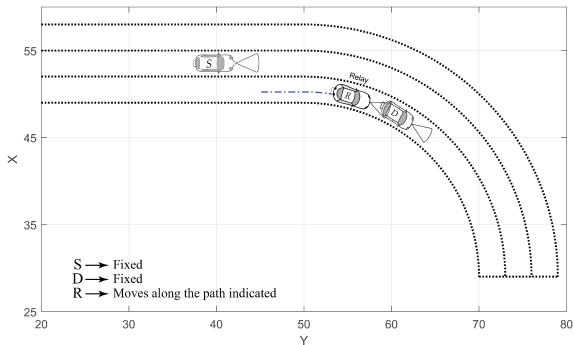


Internet of Things and Vehicular Networks (TP-546)

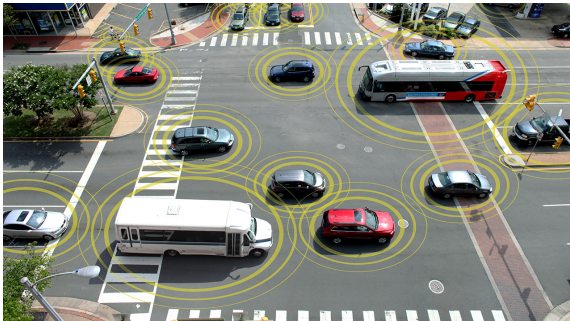
Samuel Baraldi Mafra

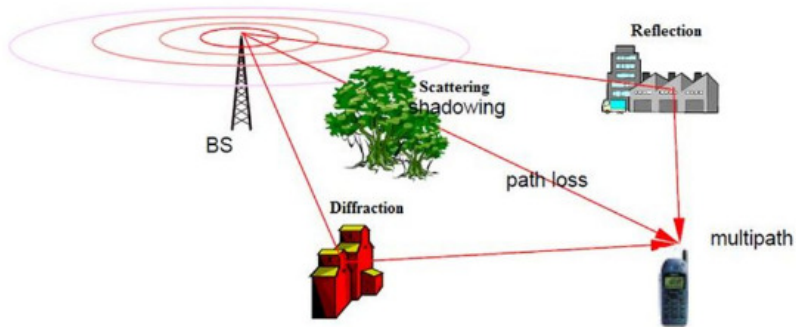


Cooperative networks



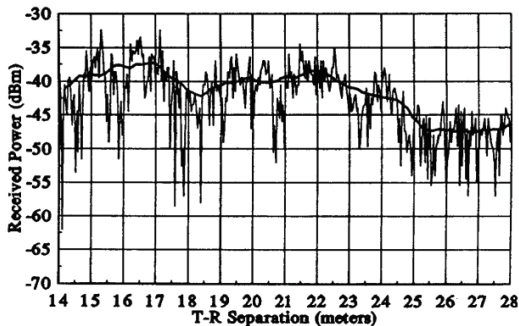
Vehicular communication allows vehicles to share information about traffic congestion, weather conditions, and accident prevention.



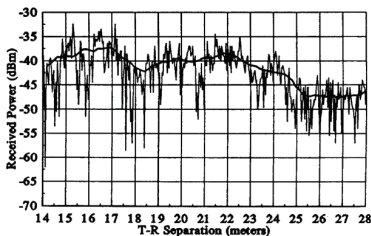


Large-Scale vs. Small-Scale

- Path loss: describes the average received power $\overline{P_r}$;
- Fading: describes the behavior of the instantaneous received power P_r .



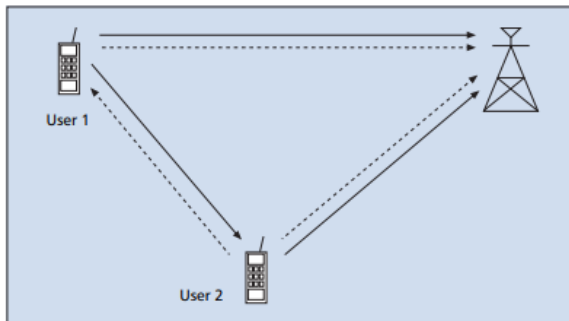
- When a mobile device moves, even over very short distances, the instantaneous signal power may fluctuate;
- This is due to multipath: the received signal is the sum (constructive/destructive) of several contributions with different phases;
- The received signal power can vary considerably.



Physical factors related to multipath:

- Presence of reflecting and scattering objects in the channel.
- Relative motion between the transmitter and receiver.
- Relative motion between objects in the channel and the transmitter and/or receiver.
- Bandwidth of the transmitted signal relative to the channel bandwidth.

- Cooperative communication involves processing information overheard by surrounding nodes and forwarding it to the destination to create spatial diversity.
- One or more relay nodes cooperate in the transmission between the source and destination.



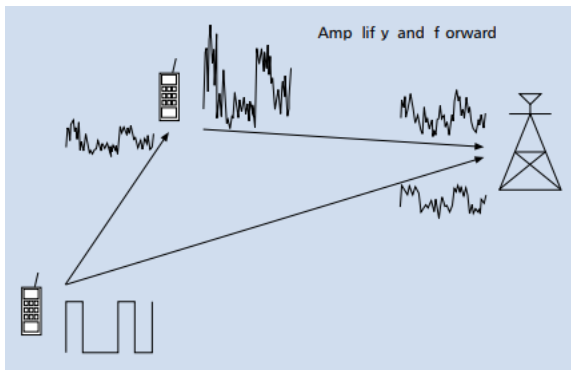
- Cooperative communication can achieve the same diversity gain as a multiple-antenna system using nodes equipped with only one antenna.
- Integrating multiple antennas into devices is difficult.

Communication protocols:

- Decode-and-forward (DF);
- Amplify-and-forward (AF).
- Store-carry-and-forward (SCF)

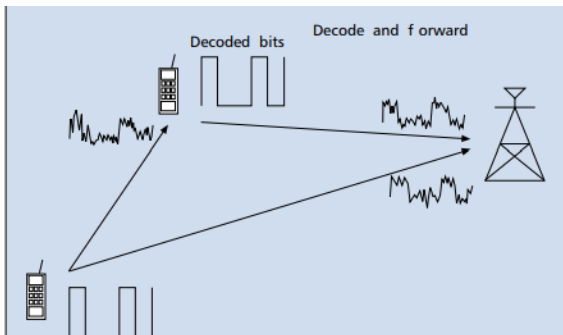
Amplify-and-forward:

- Simpler implementation;
- Both the signal and the noise are amplified.



Decode-and-forward:

- The relay node attempts to decode the message and then forwards it;
 - fixed decode-and-forward (FDF);
 - selective decode-and-forward (SDF);
 - incremental decode-and-forward (IDF).



In a vehicular network, both vehicles and infrastructure (RSUs) can act as relays:



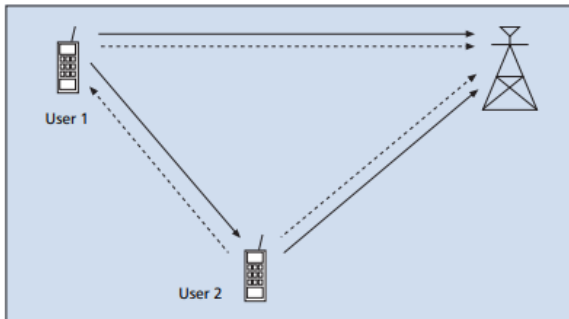
- A vehicle can assist other vehicles whose direct transmissions have failed.
- Similarly, a vehicle can help an RSU forward its packets to other vehicles outside the RSU transmission range.
- The RSU and the vehicle node participate in forwarding packets whose transmission has failed. For example, when a source RSU fails to transmit a packet successfully to the intended destination, it forwards the packet to the next RSU along the route using the wired backhaul connection. The new RSU forwards the received packet to a vehicle moving toward the intended destination. The vehicle carries and transmits the forwarded packet when it comes within transmission range of that destination.

Operating modes:

- Half-duplex;
- Full-duplex.

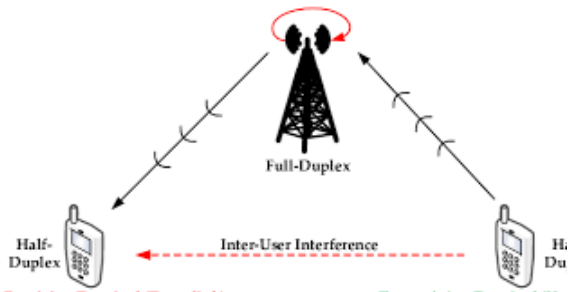
Half-duplex

- Communication takes place in two phases.
- Source-to-relay;
- Relay-to-destination.



Full-duplex

- Transmissions take place simultaneously;
- In RF systems, self-interference occurs.

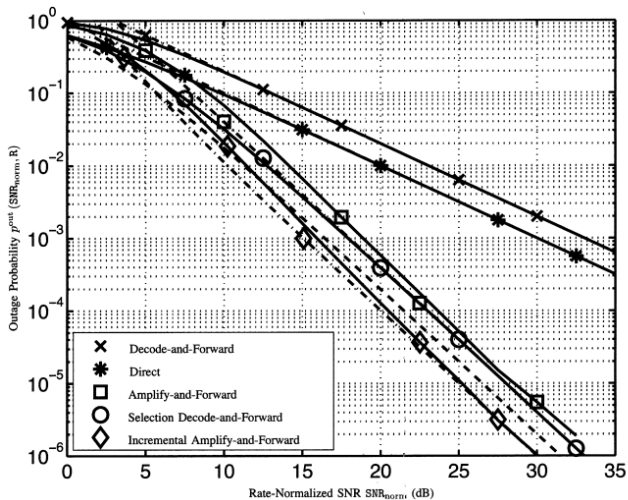


Self-interference cancellation

- Antenna isolation;
- Cancellation using analog and digital techniques.

Metrics

- Outage probability: the probability that the received signal-to-noise ratio (SNR) or the received data rate falls below a given value.



¹J. N. Laneman, D. N. C. Tse and G. W. Wornell, "Cooperative diversity in wireless networks: Efficient protocols and outage behavior," in IEEE Transactions on Information Theory, vol. 50, no. 12, pp. 3062-3080, Dec. 2004, doi: 10.1109/TIT.2004.838089.

Multiple relays

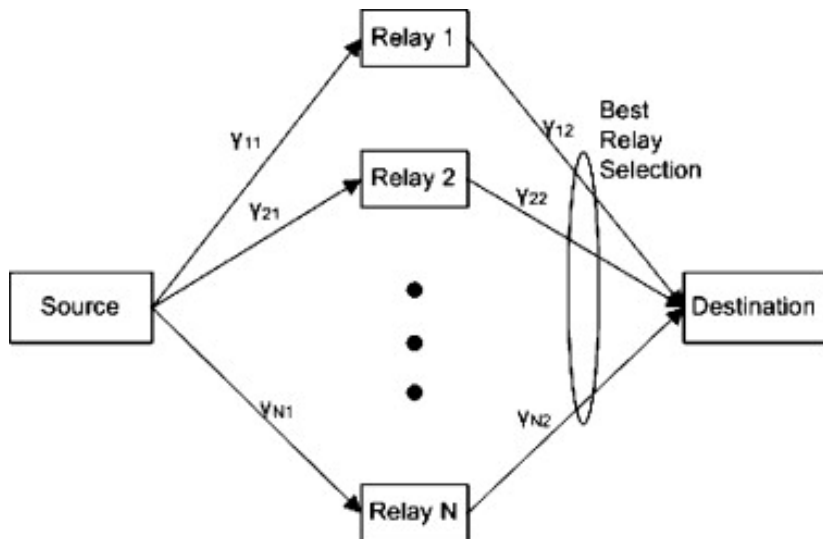
- Cooperation provides a gain even with a single relay;
- How does performance change as we increase the number of potential relays?
- How can we select the best relay node?

Relay selection:

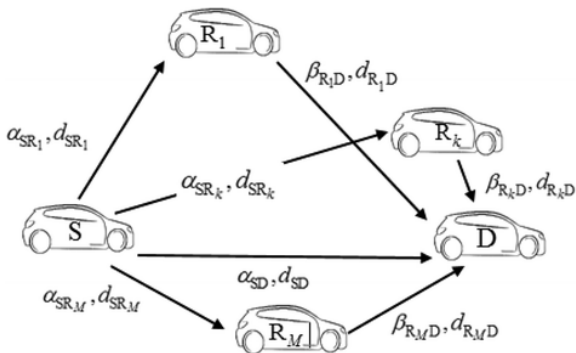
- Centralized;
- Distributed.

Relay selection:

- Proactive: the source node has channel information and selects the best relay based on this knowledge.
- Reactive: the source node sends the information to all relay nodes, and the one with the best metric forwards it.



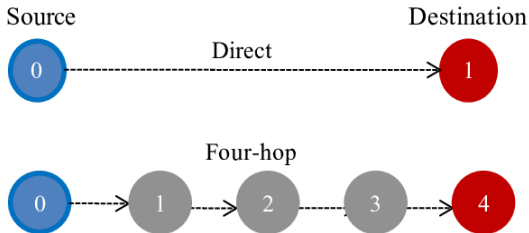
Considering only the distances, which relay is best?



Proactive:

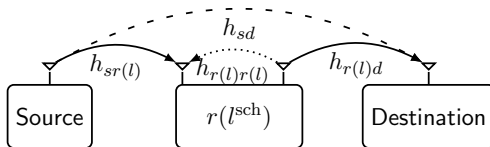
- Max-min technique: selecting the best relay means choosing the relay with the best worst instantaneous channel.
- Select the worst channel for each relay;
- Select the relay with the best metric.
- The best location is on the line halfway between the source and destination.

How is the transmission rate controlled in a multihop network?



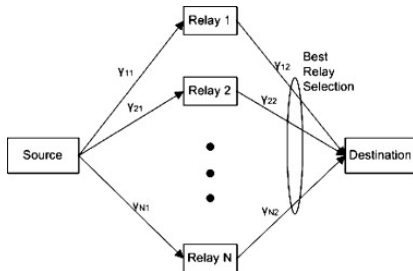
Samuel Baraldi Mafra, Evelio MG Fernandez, and Richard Demo Souza. "Performance analysis of full-duplex cooperative communication in vehicular ad-hoc networks." IFAC-PapersOnLine 49.30 (2016): 227-232.

System model:

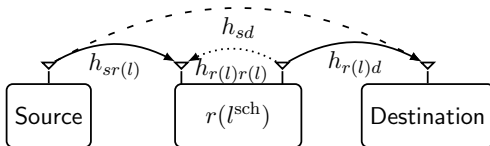
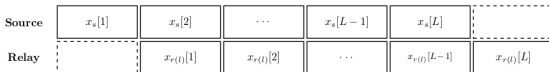


Relay selection:

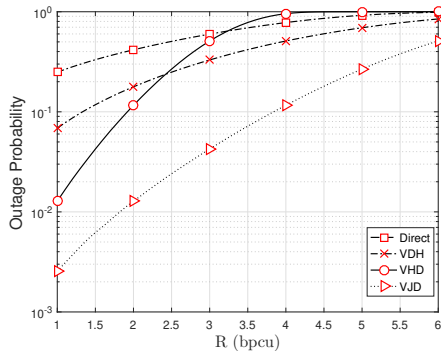
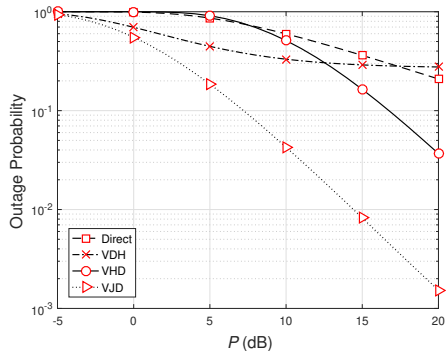
- The relays are clustered, i.e., they are all at the same distance from the destination.
- The relay with the best worst link is selected;



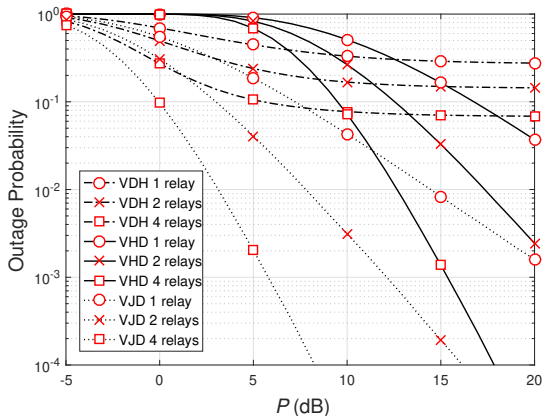
Joint packet decoding:



Outage probability as a function of P and R



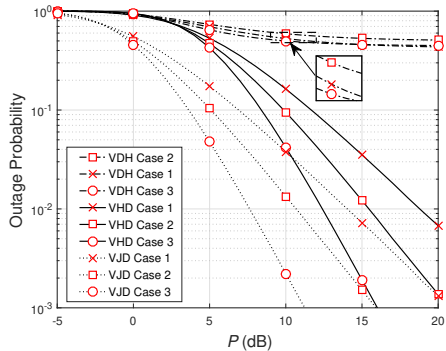
Comparison for different numbers of relays



Comparison for different channel fading parameters

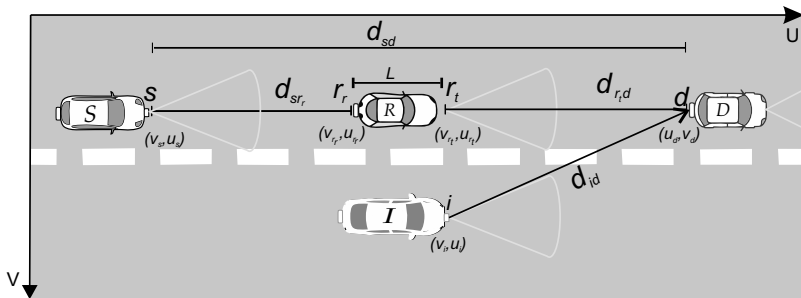
Scenarios

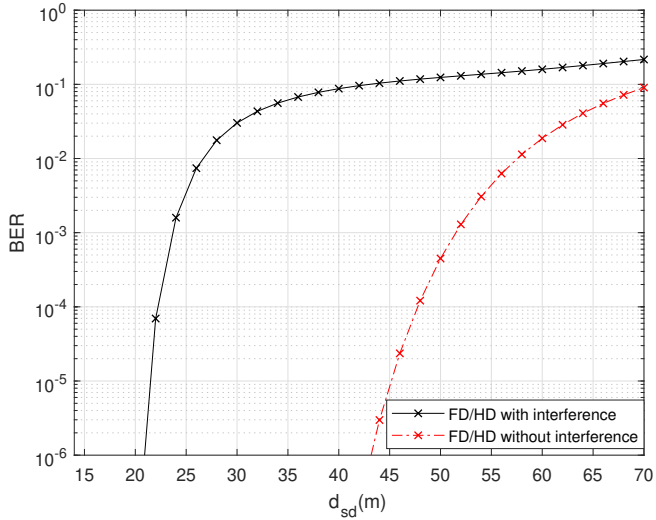
- Case 1: Sub-Rayleigh fading on the source-destination link and Rayleigh fading on the other links;
- Case 2: Rayleigh fading on all links;
- Case 3: Source-relay and relay-destination links with some line of sight, and Rayleigh fading on the other links.

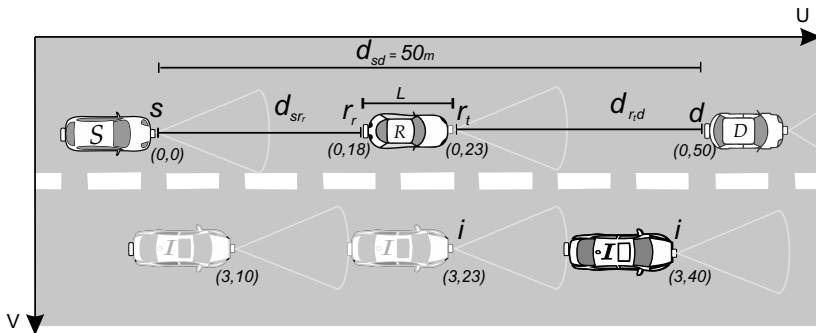


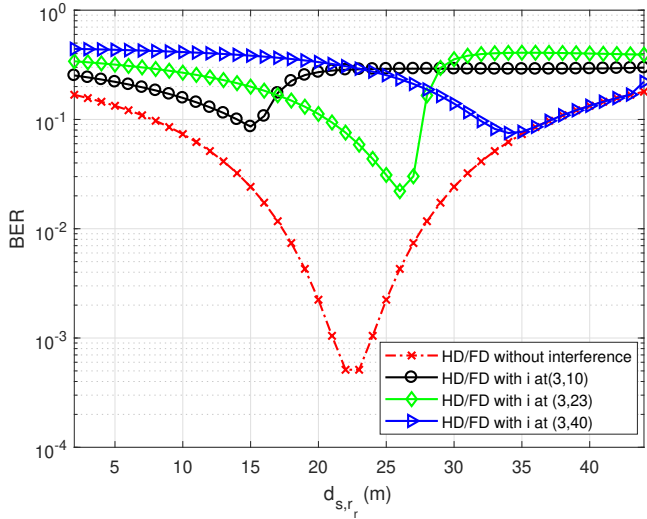
Visible light communication

Zúniga, Diego J. Cuba, et al. "Visible Light V2V Cooperative Communication Under Environmental Interference." SBrT (2019).

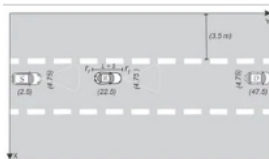




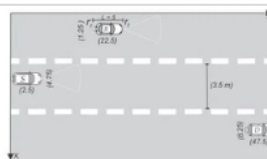




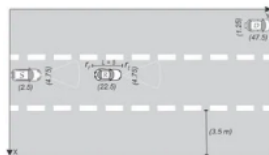
Cuba-Zúñiga, D.J.; Mafra, S.B.; Mejía-Salazar, J.R. Cooperative Full-Duplex V2V-VLC in Rectilinear and Curved Roadway Scenarios. *Sensors* 2020, 20, 3734.



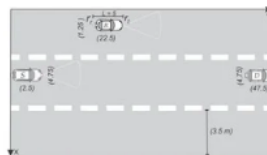
(a) Scenario A.



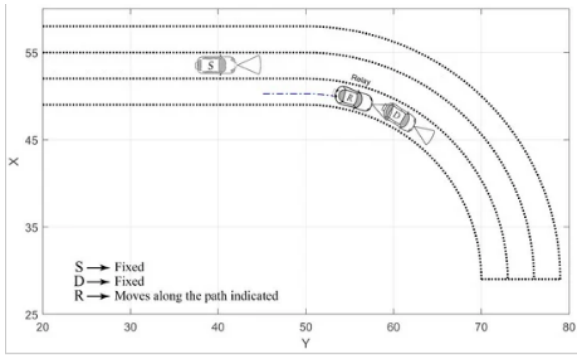
(b) Scenario B.



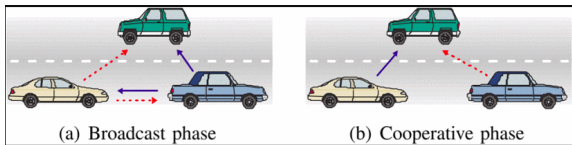
(c) Scenario C.



(d) Scenario D.

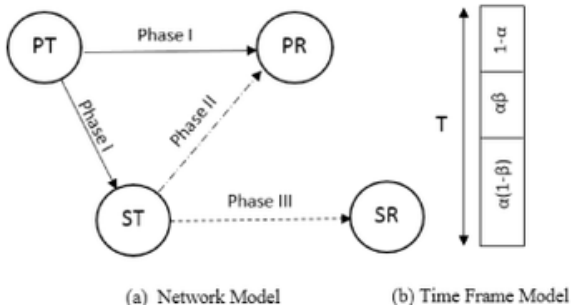


Network coding

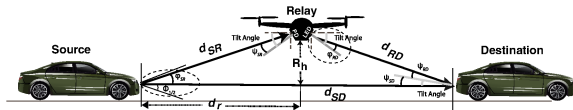


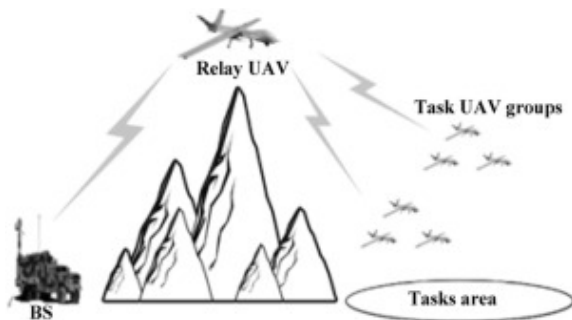
O. K. Rayel, J. L. Rebelatto, R. D. Souza and B. F. Uchoa-Filho, "Network Coding for Cooperative MIMO Vehicular Ad-Hoc Networks," 2013 IEEE 78th Vehicular Technology Conference (VTC Fall), Las Vegas, NV, 2013, pp. 1-5, doi: 10.1109/VTCFall.2013.6692301.

Cooperation in cognitive networks

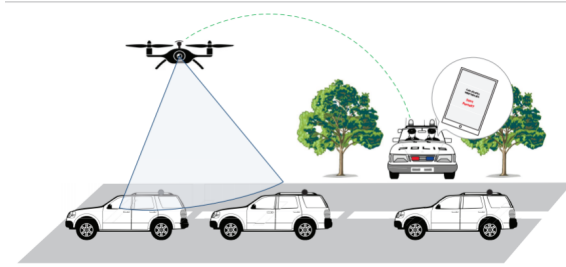


Use of drones in vehicular networks:





Vehicle monitoring



H. Menouar, et. al, UAV-Enabled Intelligent Transportation Systems for the Smart City: Applications and Challenges, IEEE Commun. Mag., Mar. 2017

